

Validating AI for Image Analysis

Part 3: Outperformance Claims & Rankings

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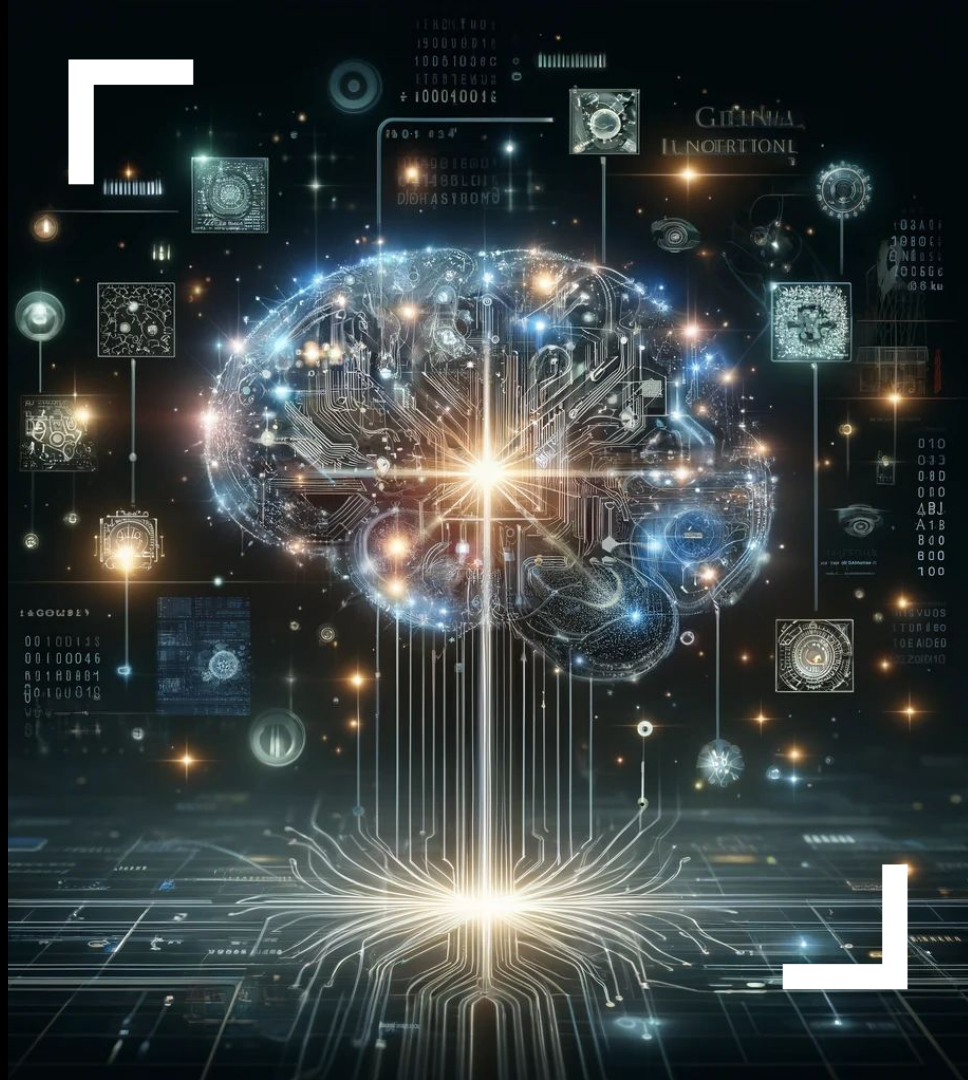
Div. Intelligent Medical Systems division, German Cancer Research Center (DKFZ)

Helmholtz Imaging

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Research for a Life without Cancer

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Agenda & General Remarks (June 25th 2026)



Time	Topic
09:00	Short Recap of Day 2
09:10	Outperformance claims + 1st practical
09:30	Rankings Part 1 + 2nd practical
10:30	Break
10:40	Teaser on more advanced ranking schemes
10:50	Open question section

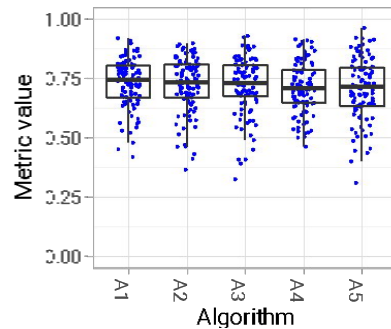
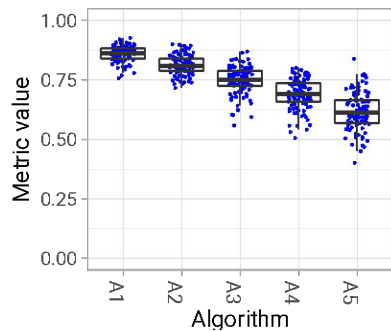
Remark 1: If not done so already, please pull from our Github repository

Remark 2: Please keep menti.com open. We will use it throughout the lecture



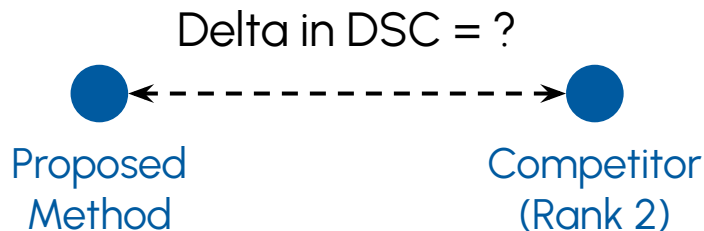
Recap from Lecture 2

Why do we need to report variability?



Algorithm	Rank
Algorithm 1	1
Algorithm 2	2
Algorithm 3	3
Algorithm 4	4
Algorithm 5	5

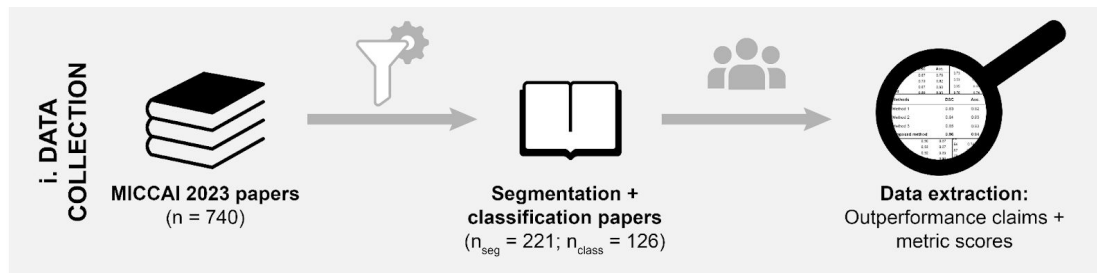
By which margin does a proposed method beat the strongest competitor?



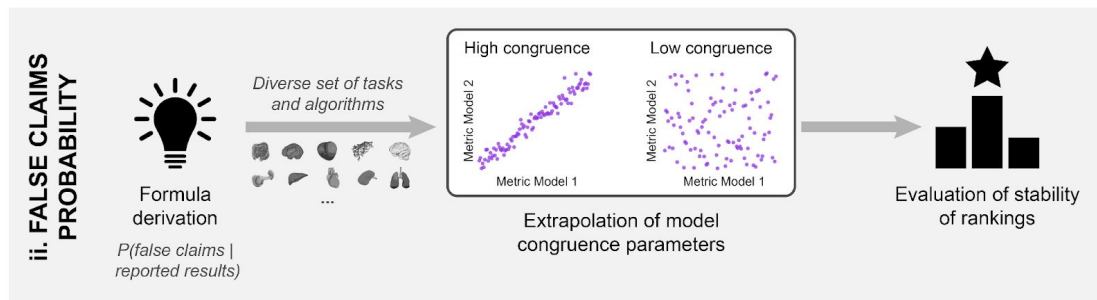
83.3% of papers claimed that they outperformed the state of the art.

Median difference in DSC: **0.01**
(all MICCAI 2023 segmentation papers)

False outperformance claims?



RQ: Are common claims of outperformance in medical imaging AI well-substantiated?



Probability of false claims



- Bayesian approach to estimate whether the relative ranking of methods is likely to have occurred by chance
- Probability that the second-ranked method (**B**) was, in fact, performing equally or better than the first-ranked method (**A**), given the results reported in the paper

$$P(p_A \leq p_B | \text{reported results}) = P(p_A \leq p_B | \hat{p}_A, \hat{p}_B)$$

A: first ranked method
B: second ranked method

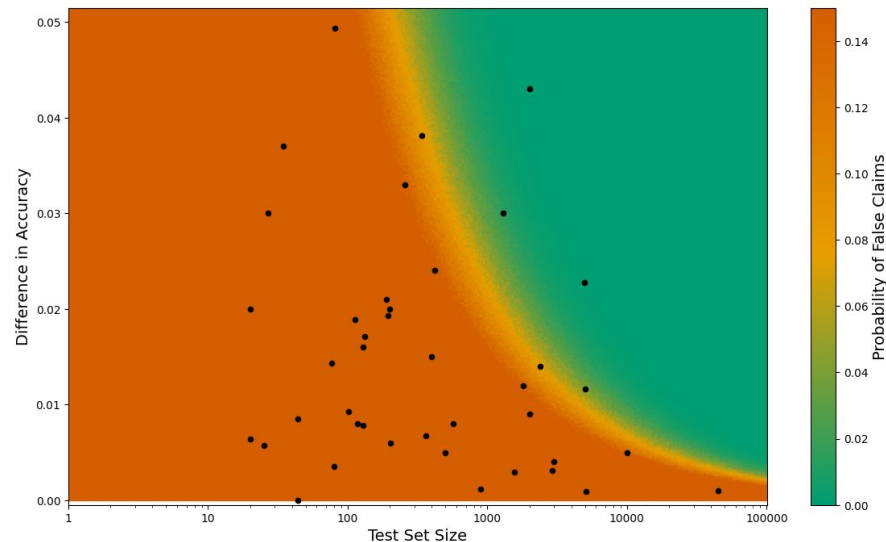
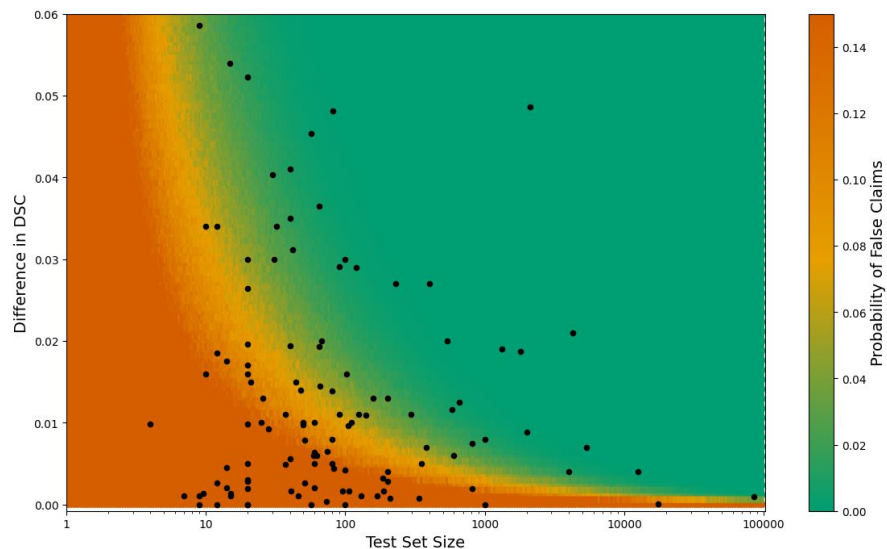
True performance (random variable)
Performance reported in the paper

Teaser: Outperformance claims are not well-substantiated



High probability ($>5\%$) of false outperformance claims in

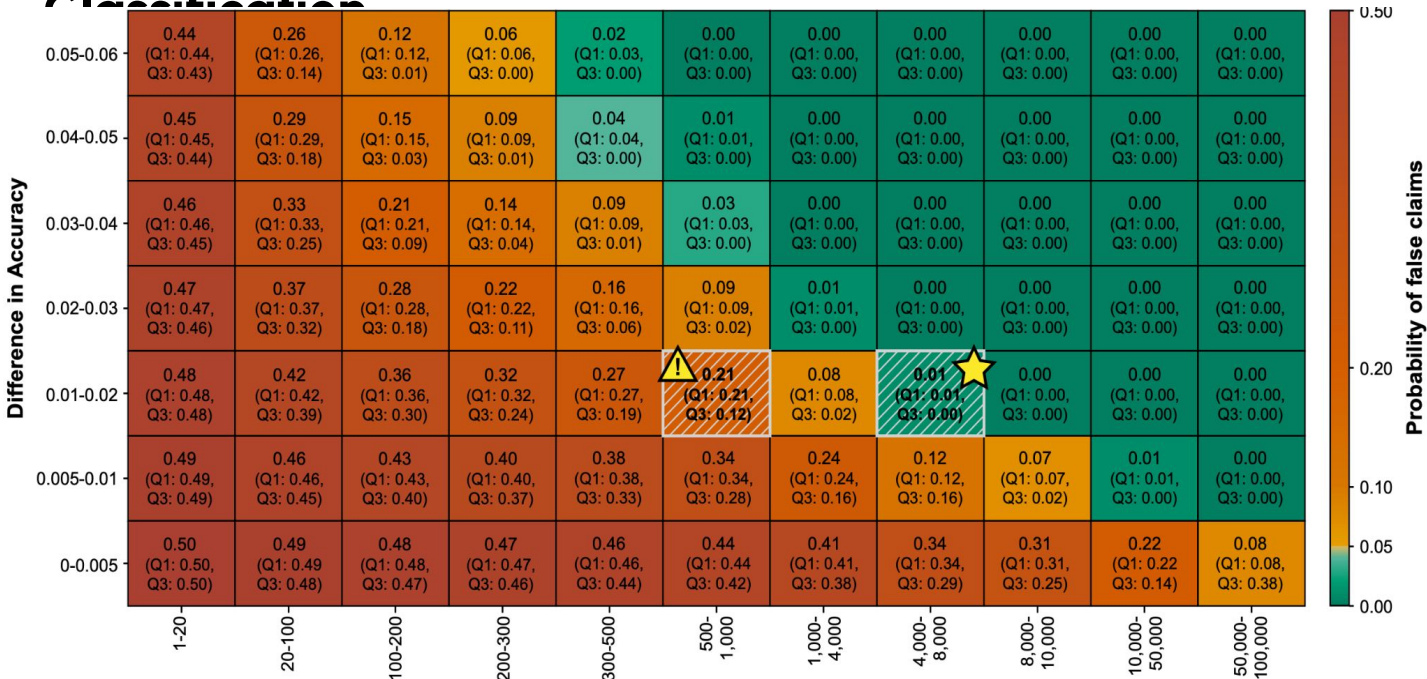
- 86% of classification papers and
- 53% of segmentation papers



Dramatically larger than usual test set sizes are needed -



Classification



8x times larger!



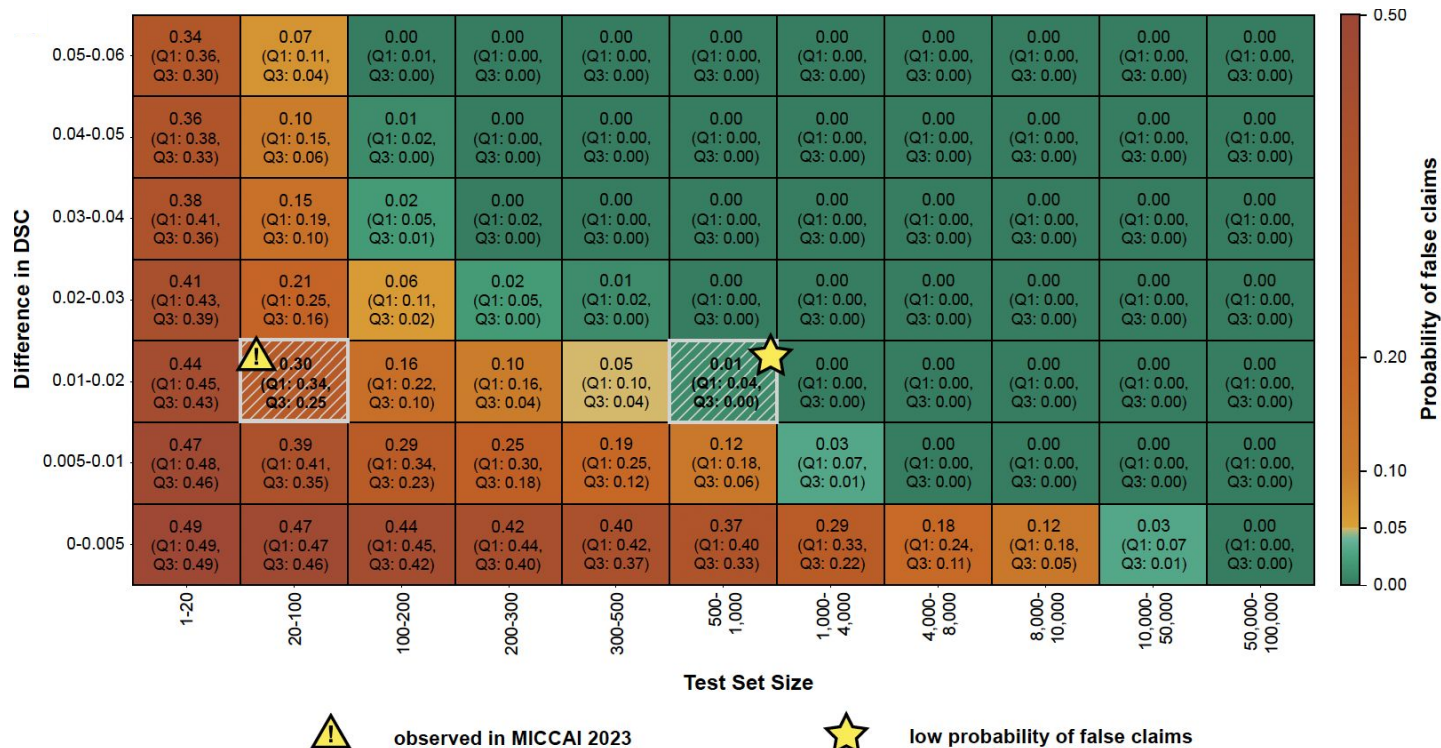
observed in MICCAI 2023



low probability of false claims

Dramatically larger than usual test set sizes are needed -

Segmentation



10x times larger!



Practical 1



Practical 1: Probability of false claims estimation



1. Go to <https://claims-reloaded.dkfz.de/> → "False claims estimator"
2. Check the boxes under "Data consent, terms of use and citation"
3. Select your scenario: "For researchers"
4. Upload the data: Go to the github repo Lecture 3 -> practical1 and pull the 4 datasets (same from Lecture 2. Start with Classification, then Segmentation)
5. Calculate the probability of false claims for each dataset
6. Compare the results with the CIs of the difference results from Lecture 2



CIs of the difference - classification



Test set, n = 20

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.050 (-0.150, 0.250)
alg_03 - alg_02	0.054 (-0.200, 0.300)
alg_03 - alg_04	0.098 (-0.150, 0.300)
alg_03 - alg_05	0.054 (-0.200, 0.300)

Test set, n = 1000

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.022 (-0.011, 0.056)
alg_03 - alg_02	0.021 (-0.012, 0.055)
alg_03 - alg_04	0.024 (-0.012, 0.058)
alg_03 - alg_05	0.018 (-0.014, 0.053)

CIs of the difference - segmentation

Test set, n = 50

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.017 (0.011, 0.023)
alg_03 - alg_02	0.012 (0.005, 0.019)
alg_03 - alg_04	0.015 (0.008, 0.022)
alg_03 - alg_05	0.009 (0.003, 0.016)

Test set, n = 5000

Comparison	Mean difference across bootstrap resamplings (95% CI)
alg_03 - alg_01	0.024 (0.023, 0.025)
alg_03 - alg_02	0.021 (0.020, 0.022)
alg_03 - alg_04	0.022 (0.021, 0.023)
alg_03 - alg_05	0.019 (0.018, 0.020)



Rankings



Rankings determine the best method



				...
	DSC: 0.87	DSC: 0.68	DSC: 0.94	...
	DSC: 0.76	DSC: 0.62	DSC: 0.81	...
	DSC: 0.90	DSC: 0.71	DSC: 0.86	...
	DSC: 0.83	DSC: 0.66	DSC: 0.92	...
...	...	...	...	...



Practical 2.1

Practical 2.1



Algorithm	Mean DSC	Median DSC	Rank (Mean)	Rank (Median)
A1	0.787	0.642	1	10
A2	0.763	0.841	2	2
A3	0.741	0.820	3	3
A5	0.729	0.857	4	1
A4	0.704	0.797	5	4
A10	0.692	0.694	6	9
A6	0.673	0.774	7	5
A9	0.668	0.696	8	8
A7	0.653	0.763	9	6
A8	0.626	0.725	10	7

Ranking robustness



RS 00	RS 01	RS 02	RS 03	RS 04	RS 05	RS 06	RS 07	RS 08	RS 09	RS 10	RS 11
DSC	DSC	DSC	DSC	HD	HD	HD	HD	HD95	HD95	HD95	HD95
Metric	Metric	Cases	Cases	Metric	Metric	Cases	Cases	Metric	Metric	Cases	Cases
Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
A1	A1	A1	A1	A5	A6	A5	A6	A6	A1	A6	A1
A2	A3	A2	A2	A6	A5	A6	A5	A5	A9	A1	A6
A3	A4	A6	A3	A4	A4	A4	A4	A3	A6	A5	A4
A4	A2	A3	A6	A11	A11	A2	A2	A1	A3	A4	A5
A5	A5	A4	A5	A2	A3	A7	A3	A2	A7	A3	A3
A6	A6	A5	A7	A3	A2	A11	A11	A4	A4	A2	A9
A7	A7	A7	A4	A10	A9	A3	A7	A11	A11	A9	A2
A8	A8	A8	A8	A7	A7	A9	A9	A7	A5	A7	A7
A9	A9	A11	A9	A9	A10	A10	A10	A9	A2	A11	A11
A10	A11	A12	A10	A8	A1	A1	A8	A10	A10	A10	A8
A11	A12	A10	A11	A1	A8	A8	A1	A8	A8	A8	A10
A12	A10	A9	A12	A12	A12	A12	A12	A12	A12	A12	A12
A13	A13	A13	A13	A13	A13	A13	A13	A13	A13	A13	A13

Kendall's Tau Rank Correlation



Coefficient

Measures agreement between two rankings

- Based on pairwise comparisons of ranked items
- **Concordant pair**: same relative order in both rankings (C: number of concordant pairs)
- **Discordant pair**: different relative order in the two rankings (D: number of discordant pairs)

$$\tau = \frac{C - D}{\binom{n}{2}}$$

 *total amount of possible pairs of observations*

$$-1 \leq \tau \leq 1$$

- +1: identical rankings
- 0: no association
- -1: reverse rankings

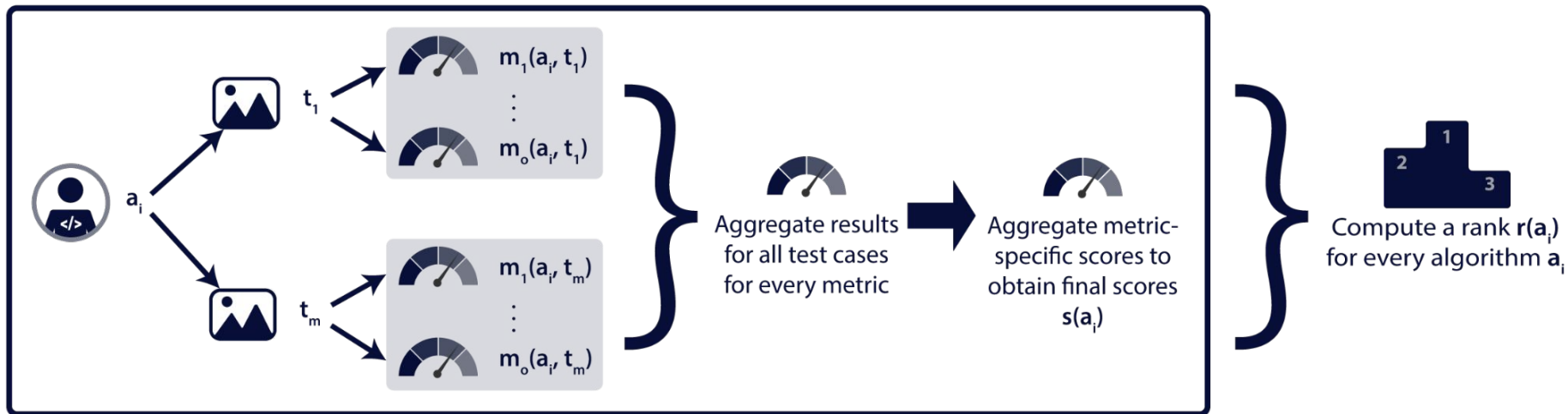
Kendall's Tau Rank Correlation Coefficient



Original	Variant 1
1	6
2	5
3	4
4	3
5	2
6	1

$\tau = -1.0$

Metric-based aggregation



- Compute a rank for each metric and algorithm by aggregating metric values over all test cases.
- If multiple metrics are used, the final rank is determined by aggregating the metric ranks

Metric-based aggregation



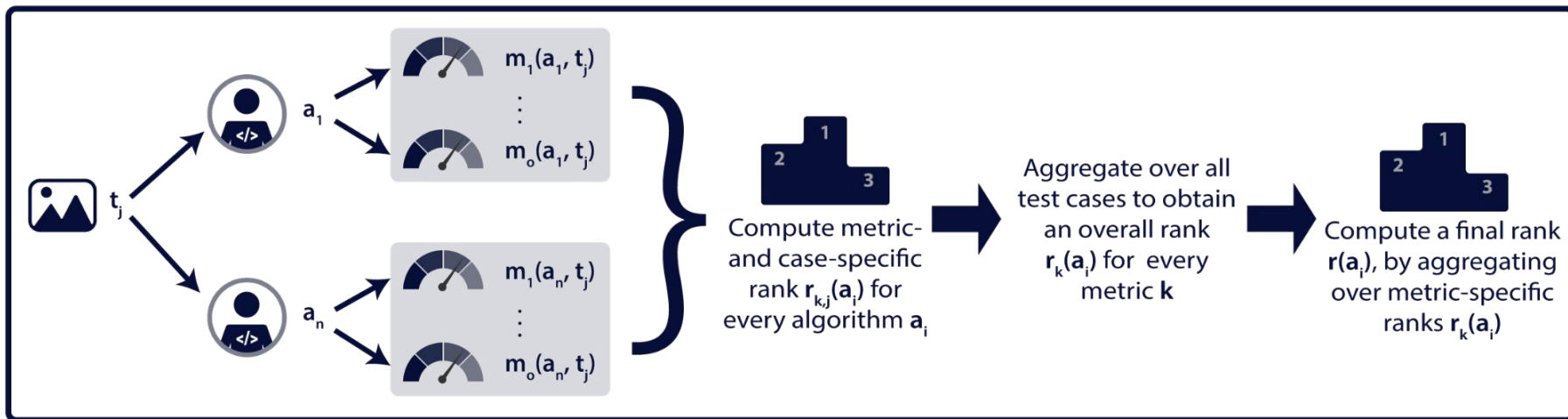
Pros

- Simple and easy to interpret
- Preserves the magnitude of performance differences
- ...

Cons

- Sensitive to the choice of aggregation function
- Missing values (NaNs) require additional handling
- When using multiple tasks, aggregation may be problematic when tasks use different metrics
- ...

Case-based aggregation



- Compute a rank for each test case and algorithm for one or multiple metrics.
- The final rank is determined by aggregating test case ranks.

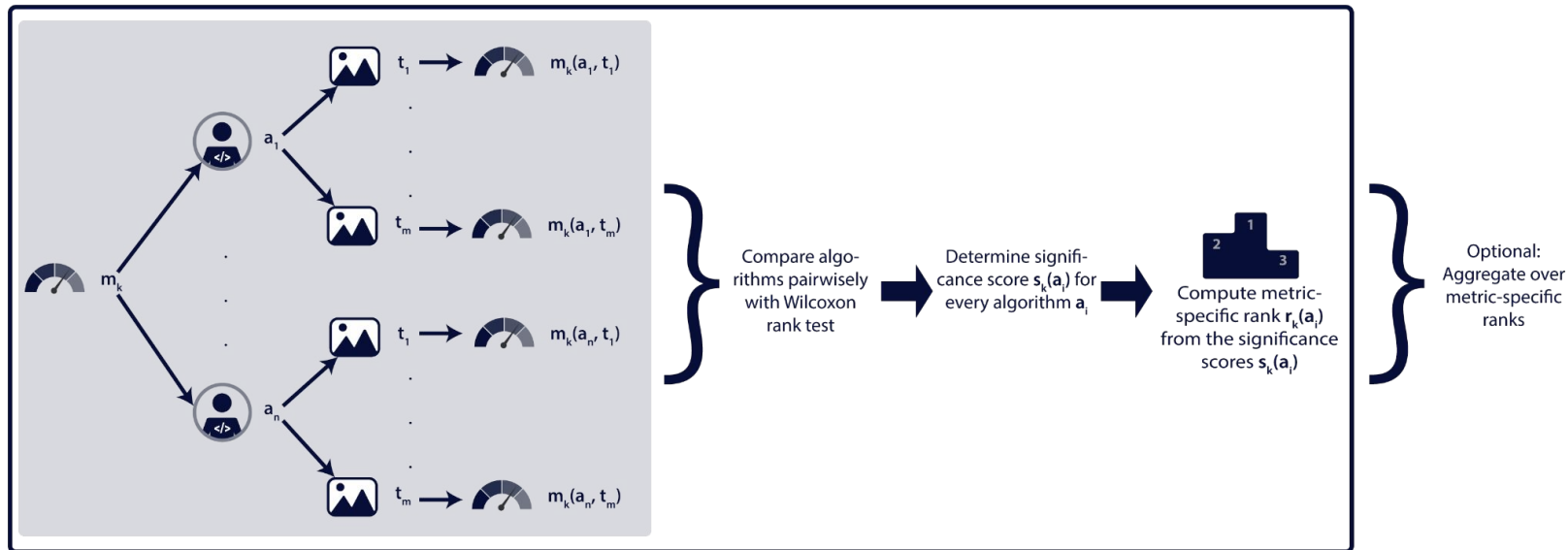
Pros

- Naturally combines multiple metrics with different ranges
- Missing values can easily be handled by assigning the worst rank for a case
- ...

Cons

- Discards information about the magnitude of performance differences
- Often produces less stable rankings than metric-based aggregation
- Rankings may change substantially after introducing additional algorithms
- ...

Significance-based ranking



- Perform all pairwise comparisons between algorithms
- Determine a significance score, i.e. the number of algorithms performing significantly worse than the current algorithm.
- Compute the ranking based on these scores.

Significance-based aggregation



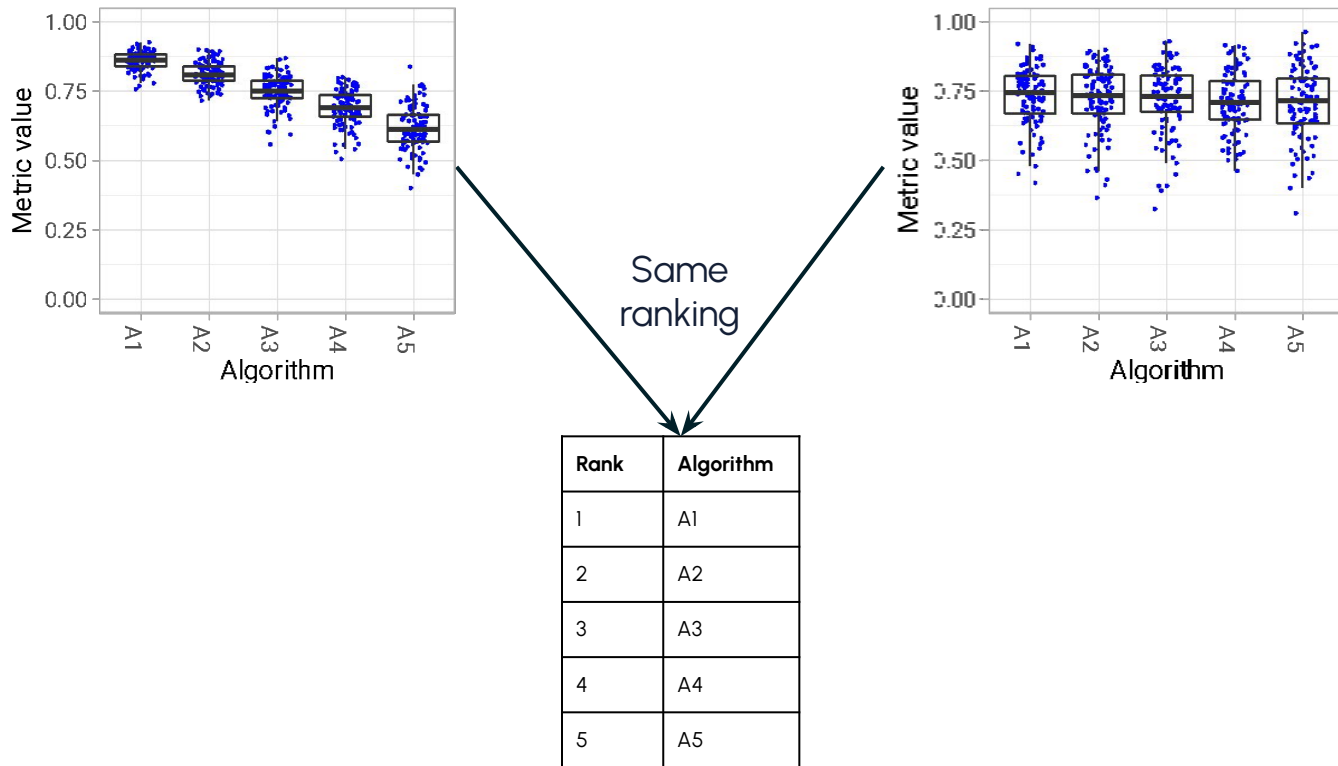
Pros

- Focuses on whether observed differences are likely to be meaningful
- Reduces the impact of negligible performance differences
- ...

Cons

- Strongly dependent on sample size
 - Small sample size: true differences may not reach significance
 - Very large sample size: negligible differences may become significant
- Rankings depend on the chosen statistical test and significance level
- A statistically significant ranking is not necessarily practically relevant
- Significance scores from different tasks may not be directly comparable
- ...

Why result visualization is important

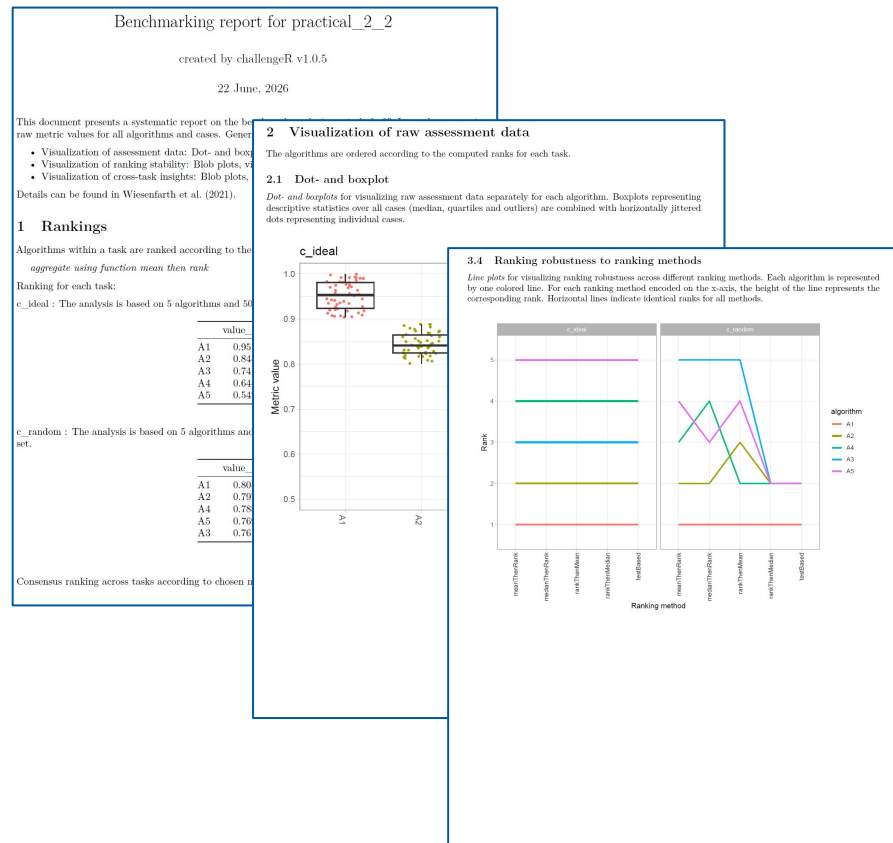


Practical 2.2

Practical 2.2: Investigate ranking robustness



- Use Rankings Reloaded: An open-source toolkit for visualizing benchmarking results
- For given data, tool produces PDF reports with ranking analyses and various visualization techniques
- Mostly for segmentation (i.e., one score per image available)
 - → the associated R tool *challengeR* can also be used for customization and for adaptation to other task types



Practical 2.2: Investigate ranking robustness

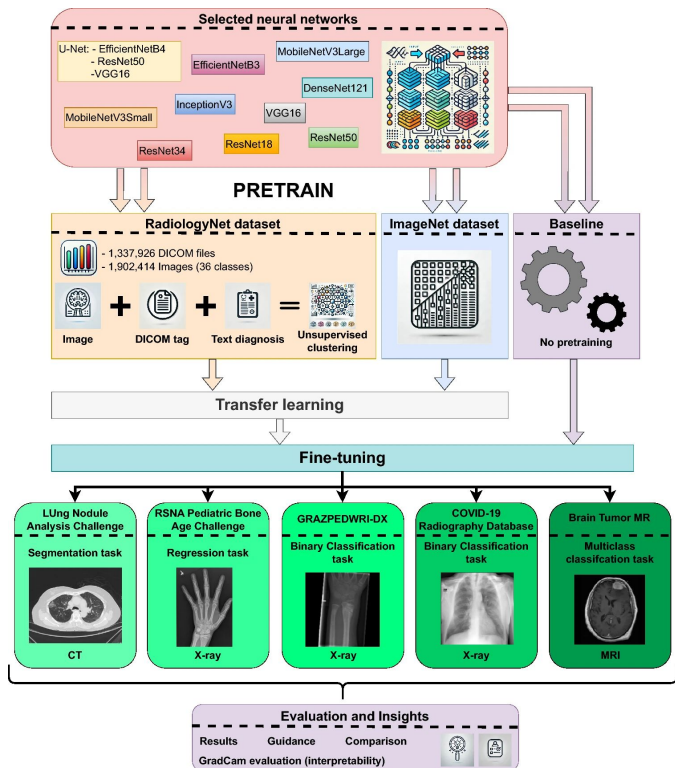


1. Open <https://www.rankings-reloaded.de/> → "Configure report"
2. Upload `data_day3_practical_2_2.csv`
3. Select:
 - a. Metric-based aggregation
 - b. Mean aggregation
 - c. 1,000 bootstrap samples
4. Generate a PDF report
5. Compare the results and plots for *c_ideal* and *c_random*
6. Which benchmark provides more robust rankings, and what evidence supports your conclusion? (*will be discussed on menti*)



More advanced ranking schemes

Foundation models: What are they?

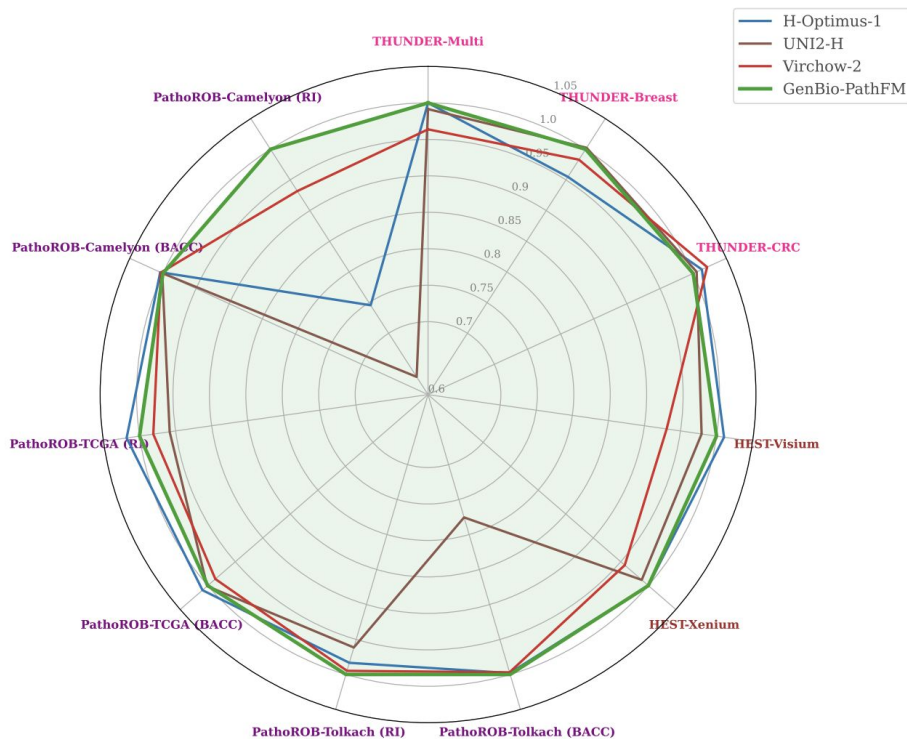


- Pre-trained on large, diverse datasets.
- Adaptable to many downstream tasks via fine-tuning or prompting.
- Provides a reusable foundation across applications and modalities.
- Scaling unlocks new capabilities but also amplifies limitations (e.g. bias and fairness, hallucinations, computational costs, safety risks).

Nappravnik, Mateja, et al. "Lessons learned from RadiologyNET foundation models for transfer learning in medical radiology." *Scientific reports* 15.1 (2025): 21622.

Bommasani, Rishi, et al. "On the opportunities and risks of foundation models." *arXiv preprint arXiv:2108.07258* (2021).

Radar plots



- Visualize multiple performance metrics simultaneously.
- Highlight model strengths and weaknesses across tasks.
- Best for capability profiling; less suitable for overall ranking.

Further interesting ranking schemes



- Erickson, Nick, et al. "Tabarena: A living benchmark for machine learning on tabular data." *Advances in Neural Information Processing Systems* 38 (2026).
- Rofin, Mark, et al. "Vote'n'rank: Revision of benchmarking with social choice theory." *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. 2023.

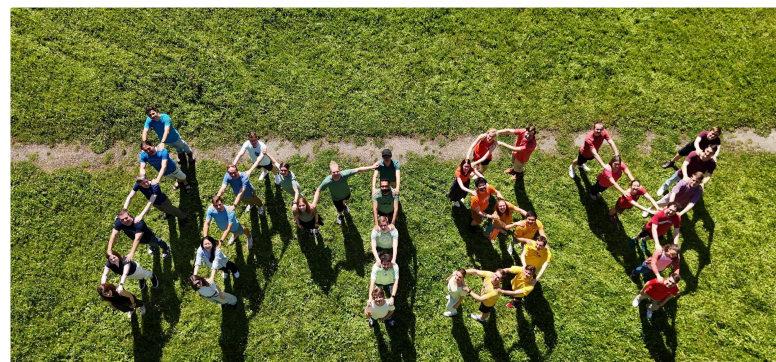
Open questions

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